PROJECT TREG

Merck & Co. / Pandion Therapeutics ($1.85bn)

Integrated Acquisition Model — Methodology & User Guide

Risk-adjusted NPV · ASC 805 purchase accounting · Accretion/Dilution · Football Field

Premium edition · Built from public SEC filings · v1.0

This guide accompanies **Merck_Pandion_Acquisition_Model.xlsx**. It explains every schedule, the methodology behind it, the sourcing of each assumption, and how to drive the model. It is written for practitioners who already understand three-statement and DCF mechanics and want the biotech-specific layer.

1. The Transaction in One Page

Item	Detail	Source
Announced / Closed	25-Feb-2021 / 1-Apr-2021	Merck & Pandion press releases
Structure	All-cash tender offer + §251(h) back-end merger	Merger Agreement
Price / Equity value	$60.00 per share / ~$1,850mm	Merck PR
Net cash / Implied EV	~$203mm / ~$1,647mm	SC 14D-9
Premium	~134% vs. 24-Feb-2021 close of $25.65	SC 14D-9
CVR / earnout	None — clean cash deal	SC 14D-9
Lead asset	PT101 — engineered IL-2 mutein / Treg agonist (post-Phase 1a)	Pandion 10-K
Advisors	Credit Suisse (Merck) · Centerview (Pandion)	Merck PR
Financing	100% Merck balance-sheet cash; no new debt	Merck FY2020 10-K

> **Why this deal is the right teaching case.** It is a textbook single-asset, pre-revenue, clinical-stage biotech acquisition: no earnings, no EBITDA, no debt, no CVR. Standard DCF and EV/EBITDA break down; the value lives entirely in **risk-adjusted pipeline NPV**. That makes it the cleanest possible vehicle for the biotech-IB toolkit.

2. Valuation Spine — Risk-Adjusted NPV (rNPV)

Banks value clinical-stage biotech with **rNPV** (also eNPV), which separates two distinct risks: scientific/clinical risk (handled by probability of success, applied to cash flows) and time value of money (handled by the discount rate). Collapsing both into one inflated discount rate is the amateur tell.

$$\text{rNPV} = \Sigma_t \left[\text{CF}_t \times \text{Cumulative PoS}_t \right] / (1 + r)^{t-0.5}$$

The model applies cumulative PoS to product revenue and post-launch costs, and weights forward development spend by the probability of *reaching* that phase. Mid-year discounting is used throughout (the `−0.5` exponent), matching banker convention.

2.1 Probability of Success

Phase-transition benchmarks (BIO/Informa/QLS, *Clinical Development Success Rates 2011–2020*, 9,704 programs; immunology-specific transitions from *J. Allergy Clin. Immunol.* 2017):

Transition	All indications	Immunology	Model base (post-Ph1a, biomarker-confirmed)
Phase 1 → 2	~52%	65.7%	(passed — Ph1a complete)
Phase 2 → 3	~29%	30.8%	55%
Phase 3 → NDA	~58%	64.9%	70%
NDA → Approval	~89%	89.2%	90%
Cumulative	**~7.9%**	**~12%**	**~35%**

> **On the base-case PoS uplift.** The naive industry-average cumulative PoS for a Phase-1 biologic is ~9%. The model's ~35% base reflects three defensible adjustments: (i) Pandion had *completed* Phase 1a with positive Treg-selectivity biomarker data, removing the single largest attrition gate; (ii) Centerview's fairness DCF relied on management projections that embed materially higher implied PoS; and (iii) Treg agonism is a mechanistically validated pathway (cf. Lilly/Nektar NKTR-358). The Downside scenario reverts to ~19% to show the bear case. **Every PoS input is a blue cell in `Drivers` — change it.**

2.2 Discount Rates — and the Acquirer Bridge

- **Standalone clinical biotech WACC: 12.0–14.0%** — the exact band Centerview used in its fairness DCF.
- **Acquirer (Merck) corporate WACC: ~8.5%** with 0% terminal growth — the published bank standard for modelling Merck itself.

This gap is the heart of the deal. The same risk-adjusted cash flows are worth far more to Merck (8.5%) than to a standalone biotech investor (13%). The model makes this explicit in the **Acquirer-WACC Strategic Revaluation** block on `Valuation_Summary`: standalone rNPV anchors the floor; revaluation at the acquirer's cost of capital plus a control/scarcity premium bridges to $60.00.

3. Schedule-by-Schedule Walkthrough

Tab	What it does	Key mechanic
Drivers	Single control panel. All blue inputs + 3-way scenario switch (C6)	CHOOSE() on scenario
rNPV_PT101_UC	Full risk-adjusted DCF, lead indication, 2021–2042	PoS × revenue; phase-weighted cost; mid-year DF; −60% terminal decay

rNPV_Pipeline	SLE, PT002, PT627, PT001, Astellas royalty	Unit peak-sales NPV factor × PoS × peak × econ. share
Valuation_Summary	Sum-of-parts equity bridge + acquirer revaluation	rNPV + cash + NOL − dilution ÷ TSM shares
Trading_Comps	Centerview's 8-company set, EV-based	Reference EV + net cash ÷ shares
Precedent_Txns	Centerview's 6 precedents + extended context	Reference txn value + cash ÷ shares
Premium_Paid	Offer vs. undisturbed / VWAP / 52-wk	Premium grid
Football_Field	All methods + strategic + $60 offer line	Stacked-bar range chart
Sources_Uses	Transaction funding	Cash-only; sources = uses check
PPA	ASC 805 allocation, biz-combo / asset-acq toggle	Goodwill or Day-1 IPR&D expense
Accretion_Dilution	Merck EPS impact, GAAP & non-GAAP	Foregone interest + R&D burn + IPR&D
Pro_Forma	Combined IS + balance-sheet bridge	Merck + Pandion + adjustments; A=L+E check
Synergies	Pipeline acceleration (not cost cuts)	PoS / peak / timing dials
Sensitivity	WACC×Peak and PoS×Peak grids	Transparent proxy reconciling to live model
Scenario_Summary	Live read-out of active scenario	Reads Drivers!C6
Error_Checks	8 integrity checks + master flag	Sources=Uses, PPA ties, BS balances, PoS≤100%…

4. Purchase Accounting (ASC 805)

The model carries a toggle in `Drivers` (**1 = Business Combination, 2 = Asset Acquisition**):

- **Business combination** — acquired IPR&D is *capitalised* as an indefinite-lived intangible; the residual is goodwill; a deferred tax liability is booked on the intangible step-up (it has $0 tax basis), which increases goodwill. Subsequent R&D is expensed.
- **Asset acquisition** — if substantially all value sits in one IPR&D asset with no alternative future use, it is *expensed immediately* (the classic pharma "IPR&D charge"), no goodwill. This is the Lilly/Nektar 2017 precedent.

Illustrative business-combination allocation at base inputs: $1,872mm consideration − $203mm cash − $1,500mm IPR&D − $50mm other + $315mm DTL = goodwill plug. Both treatments flow to Accretion/Dilution and the Pro Forma balance sheet automatically.

5. Accretion / Dilution Logic

The target is pre-revenue, so there is no clean near-term accretion — and that is the point. The model quantifies what an acquirer actually communicates to the market:

- Year-1 dilution = after-tax foregone interest on $1.85bn cash + inherited Pandion R&D burn.
- One-time IPR&D charge (asset-acq treatment) hits GAAP EPS, excluded from non-GAAP guidance.
- Accretive only on successful PT101 launch (~2028).

> Standard market language, reproduced in the model's commentary cell: *"modestly dilutive to GAAP and non-GAAP EPS through clinical development; accretive upon successful PT101 launch in approximately 2028."*

6. Driving the Model

1. Open `Drivers`. Set **C6** = 1 (Base), 2 (Upside) or 3 (Downside).
2. Every blue/gold-shaded cell is an input you can change. Black = formula, green = cross-sheet link.
3. Read outputs on `Valuation_Summary`, `Football_Field`, `Scenario_Summary`.
4. Toggle accounting treatment in `Drivers` (PPA section) and watch Accretion/Dilution + Pro Forma move.
5. Confirm `Error_Checks` master flag reads **ALL OK** before relying on any output.
6. After changing inputs in Excel, allow auto-calc (or press F9 / Cmd-=) to refresh.

Scenario	Cum. PoS	rNPV PT101 UC	Value / share	Strategic / share
Base	~35%	~$640mm	~$40	~$47
Upside	~49%	~$1,490mm	~$73	~$84
Downside	~19%	~$30mm	~$14	~$24

The $60.00 offer sits above the Base standalone range and inside the Upside — precisely the real-world read: Centerview's DCF was the highest of the three methods and bracketed $60 at its upper end.

7. Sources & Caveats

- **Primary:** Pandion S-1, 10-K (FY2020), SC 14D-9 (4-Mar-2021, Centerview opinion); Merck FY2020 10-K and deal press releases; BIO/Informa/QLS success-rate study; *J. Allergy Clin. Immunol.* 2017.
- Centerview's stated implied ranges: Trading Comps $23.45–$39.40; Precedents $31.40–$55.35; DCF was the highest method and bracketed $60.
- Management PoS percentages were embedded (not separately disclosed); the model exposes them as inputs and the base case is calibrated to reconcile to Centerview's DCF zone.
- Pro Forma uses illustrative Merck consolidated approximations to teach mechanics — replace with audited detail for transaction work.
- VWAP / 52-week reference prices on Premium_Paid are illustrative; undisturbed close and IPO price are reported.

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